

Operator Discovery Report: acceptance_threshold_worse_moves_with_stagnation_bonus

- Operator type: acceptance.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Allow a small amount of bounded worse 2-opt moves using a low threshold; add extra acceptance probability when the search is stagnating to escape NN traps without turning into a full restart strategy.
- Novelty classification: threshold_acceptance.
- Rediscovery signature: acceptance:threshold:thr=0.018:temp=0.000.
- Surviving candidate: False.
- Problem class where it helps: none confirmed.

Pseudocode

- Use threshold acceptance when a restart solution is not immediately improving.
- Relax or tighten acceptance depending on stagnation and late-search cooling.
- Reject candidate restarts that exceed the configured worse-move budget.

Complexity

- Non-empty lines: 11.
- Complexity score: 0.42.
- Fingerprint: 8221967b485bffa9.

Why It Should Help

- The operator only changes restart acceptance, so any gain is attributable to a narrow exploration-vs-stability decision rule.

Failure Cases

- Does not transplant cleanly into clustered_local_search.
- No validation family shows a clear improvement over the host baseline.
- Disabling the operator does not hurt, so the ablation does not support a real operator effect.

Transplant Results

- nearest_neighbor_2opt gap delta 0.0, runtime delta -86.828329, distance-eval delta 0.0.
- cheapest_insertion_2opt gap delta 0.0, runtime delta 1034.308557, distance-eval delta 0.0.
- random_restart_2opt gap delta 0.0, runtime delta -6.184414, distance-eval delta -524.0.
- sparse_three_opt gap delta 0.0, runtime delta -1.237557, distance-eval delta 0.0.
- clustered_local_search gap delta 0.007599, runtime delta 66.128943, distance-eval delta 1329.428571.

Instance-Family Results

- heldout_tsplib operator gap 0.235058 vs baseline 0.235058.

- clustered_tsp operator gap 0.110853 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.109057 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: 0.0.
- Runtime inflation: -0.00642.
- Distance-evaluation inflation: 0.0.
- Same-gap-faster flag: True.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.0.
- improving_only_acceptance mean gap 0.121317 and delta vs full operator 0.0.
- tight_threshold_acceptance mean gap 0.121317 and delta vs full operator 0.0.